



Amazon's 'Project Como'

Amazon could be planning to launch convenience stores to sell perishable products under its "Project Como" <http://digit.in/AmzComo>

No swiping on Hinge now

Hinge is rebranding itself to position as a 'relationship app' and now costs \$7 <http://digit.in/RebrHinge>

entire stretches can be replaced with the correct DNA. When the cell's repair machinery joins the DNA back on its own, the sequence is just slightly altered, which is sufficient in a lot of cases. For example, in the case of faulty genes, single letter changes are known to inactivate them. If, however, at the cut site, a repair DNA sequence is provided, the machinery can paste that into the genome. This is useful when we need to add a new piece of DNA, or replace a faulty one. Therefore, not only can we deliver entire fragments of

genetic basis of diseases, among other things. Recently, another landmark was reached, when researchers modified the gene responsible for B thalassemia in human embryos for the first time. Meanwhile, leading CRISPR scientists ventured into the industry by forming a company, Editas Medicine, which aims to develop this technology further to treat diseases.

Our distant evolutionary cousins, the crab-eating monkeys have been a target of CRISPR technology. By changing some genes in these monkeys, they show autism like symptoms which give researchers a primate model organism for autism. In the future this route can be tried for various diseases that affect humans and because monkeys are genetically more similar to humans than lab rats and guinea pigs, the study would be easier to adapt for case of our species.

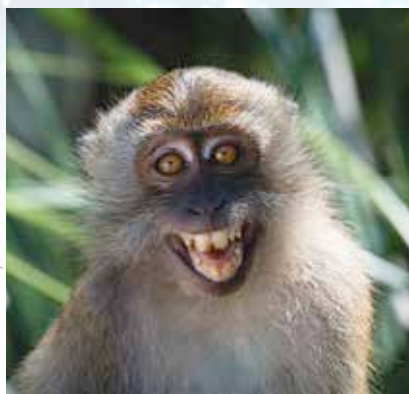
The transforming power of CRISPR is not limited to medicine, with a lot of projects in the pipeline for the use of CRISPR in the farm too. Chinese scientists have modified a wheat strain to make it resistant to fungus, and around the globe CRISPR is being utilised on crops as well as animals to make them better yielding, more resistant to infections and hostile environmental conditions.

This fervent atmosphere created in the research and biomedical industry bodes well for mankind and speaks volumes about the faith the scientific community has on CRISPR technology.

Human Trials and the quest to cure Cancer

The announcement made by NIH, allowing human trials in June this year is a shot in the arm for the scientific and medicine fields. The team at University of Pennsylvania, ironically the place where the death of Jesse Gelsinger halted gene therapy for a long time, is all poised to launch the trial on people suffering from specific forms of cancer. This trial is unique because the CRISPR/Cas9 system enables editing to be done at three different sites in the DNA (Deoxyribonucleic Acid).

How are they planning to establish a cure for Cancer? Special cells in the blood which help fight off infections, called T cells, will be taken from the patient's body. Through CRISPR/Cas9 they will be engineered in such a way that they can specifically home in on cancer cells. Also, cancer plays another trick on the T cells that reduces the strength of our body's immune response. The other two modifications in the genome are aimed at disabling the gene responsible for this immune reduction. The engineered cells will then be put back into the patient. Collectively, these modifications hold the promise to cure some forms of cancer. An approval from the U.S Food and Drug Administration is the only hurdle on this path, but the scientists hope to get the green light and start the trial by the end of this year.

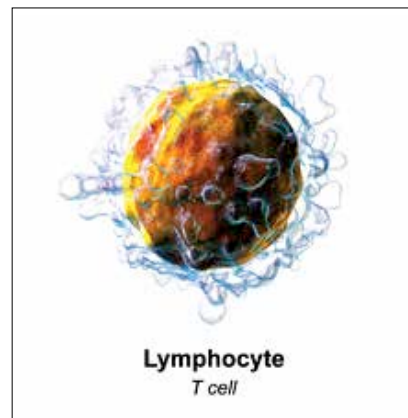


These species have been used for autism research enabled by CRISPR

gene DNA but also edit small portions in the gene with ease. Further, more than one region in the genome can be edited, providing a fix to disorders where a simple edit at one location doesn't solve the problem. CRISPR is an incredibly flexible technology in that sense and compared to earlier gene therapy techniques it offers a wider range of solutions with greater efficiency. The versatility with which this system can edit genes has caught everybody's attention. Thanks to its affordability and simplicity, a revolution is underway, with CRISPR being tried by researchers everywhere – in academia and industry.

CRISPR in action

CRISPR has been successfully used to alter genes in bacteria, yeast, fruit fly, mice, monkeys and even human cells in a petri-dish. The technique will enable scientists to directly modify genes, making it easier to explore the



A 3D reconstruction of a T-cell lymphocyte

Chinese researchers are also set to start their own CRISPR/Cas9 trial to treat patients with lung cancer. Editas Medicine is not far behind, with a proposal to use the technique to cure a form of blindness.

CRISPR is arguably the most revolutionary technology of our generation. A lot of Nobel Prizes are waiting to be won. The age of designer babies is nigh, and we can only hope for common sense to prevail. Getting carried away with genetics isn't ideal. Didn't the Nutty Professor series of movies teach you anything?